B.Sc. Hons. (CA&IT) Sem-I

U11A1IP1-INTRODUCTION TO PROGRAMMING-I

UNIT-1

Fundamental of Algorithms

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❖ What is an Algorithm?

- A step-by-step set of instructions to solve a problem or perform a task.
- A sequence of activities to be processed for getting desired output from a given input.
- Example:
  Making tea, solving a math problem, using a GPS.

➢Rules for Making an Algorithm

1. Clear Objective – Know what the algorithm is supposed to do.
2. Step-by-Step Instructions – List actions in the correct order.
3. Simple and Precise Language – Easy to read and understand.
4. Definite Start and End – Algorithm must begin and stop clearly.
5. Input and Output Defined – What data goes in and what result comes out.
6. Logical and Finite – Must follow logic and finish in a limited number of steps.

➢Properties (characteristics) of an Algorithm:

- It should be simple.
- It may accept zero or more inputs.
- It should have finite number of steps.
- It should produce at least one output.

➔ Algorithm: Sum of Two Numbers taken as input from the user

1. Start
2. Take input of the first number → A
3. Take input of the second number → B
4. Add the two numbers → Sum = A + B
5. Display the result → Print Sum
6. End

- ➢ **While writing algorithms we will use following symbol for different operations:**

Arithmetic Operators:-

‘+’ for Addition

‘-’ for Subtraction

‘*’ for Multiplication

‘/’ bs for Division

‘%’ for modulo (return the reminder after division)

Relational Operators:

‘>’ for Greater than

‘<’ for Less than

‘>=’ for Greater than equal to

‘<=’ for Less than equal to

‘=’ for equal to

‘!=’ for not equal to

Logical Operators:

‘AND’

‘OR’

‘NOT’

❖ Types of Algorithms

1. Sequence Algorithm
2. Branching (Selection) Algorithm
3. Loop Algorithm

1) Sequence Algorithm

- Steps are followed one after the other in order.
- No conditions, no repetitions — just straight steps.

Sum of Two Numbers

Step 0: Start

Step 1: Input first number into variable a

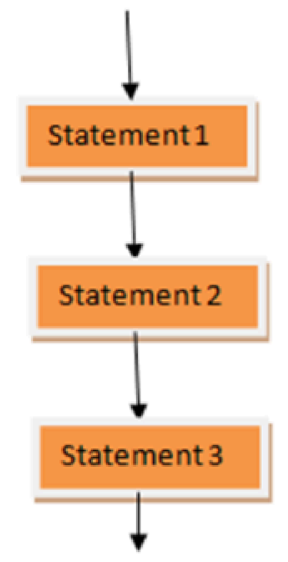
Step 2: Input second number into variable b

Step 3: Calculate sum = a + b

Step 4: Print the sum

Step 5: Stop

Sequential: The statements execute from top to bottom sequentially.



2) Branching (Selection) Algorithm/Conditional Control:

- Uses conditions (like if, else) to make decisions.
- The next step depends on a condition being true or false.

Check if a Number is Even or Odd

Step 0: Start

Step 1: Input a number into variable num

Step 2: If num % 2 == 0 then

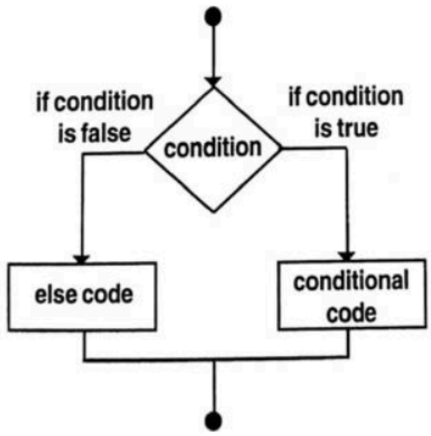
Print "Number is Even"

Else

Print "Number is Odd"

Step 3: Stop

- **Conditional or selection:** Out of two instructions, only one will be executed successfully depending on the condition. Because the condition produces a result as either True or False.



3) Loop Algorithm

- The loop (repetition) structure is the construct where statements can be executed repeatedly until a condition evaluates to TRUE or FALSE. There are three repetition structures in C:

1. WHILE
2. DO – WHILE
3. FOR

<u>Print Numbers from 1 to 5</u>

Step 0: Start

Step 1: Initialize variable i = 1

Step 2: Loop start:-Repeat steps 3 to 4 while i <= 5

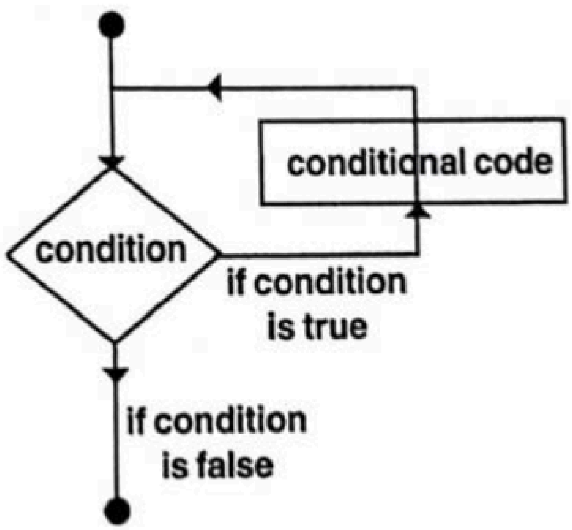
Step 3: Print the value of i

Step 4: Increment i by 1

Step 5: End loop

Step 6: Stop

- **Repetition or loop:** In which the instruction(s) are repeated over a whole list whenever the condition is **True**.



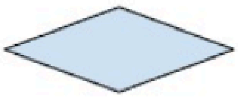
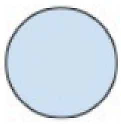
❖ Flowchart

➢What is a Flowchart?

A flowchart is a **visual diagram** that shows the steps of an algorithm or process using different shapes and arrows.

It helps to:

- Understand the logic of a program or process easily
- Plan before writing actual code
- Communicate how a task will be executed step-by-step

Symbol	Name	Function
	Start/End	Indicates the start or end point of the process.
	Process	Describes a specific action or task to be performed within the process.
	Decision	Denotes a question or condition to be evaluated. Also, the different output options based on the answer.
	Connector	Used to connect different parts of the diagram that are in separate places.
	Flow arrow	It represents the direction of the process flow, indicating the order in which actions are carried out.
	Input/Output	It is the material or data entering or leaving the system.

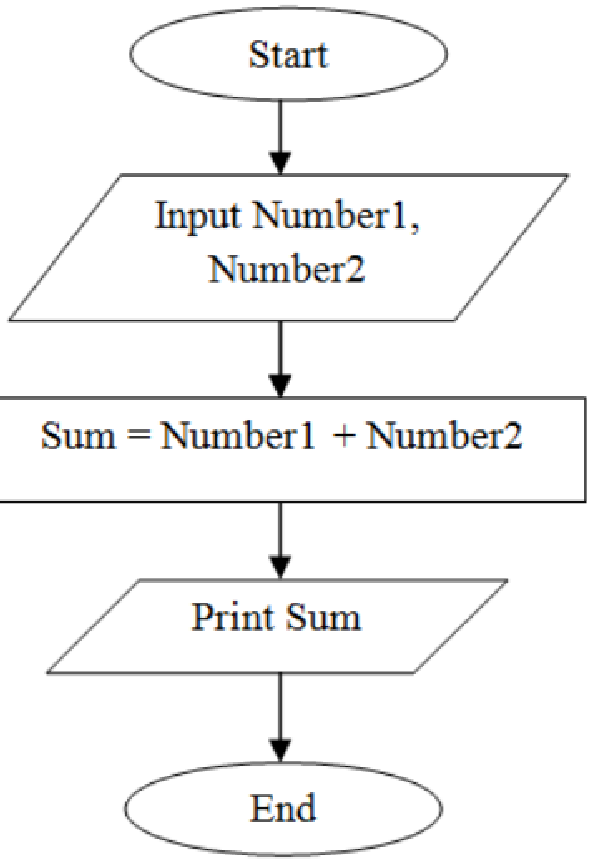
➢ Rules for Making Flowcharts:

1. Use standard symbols – Oval (Start/End), Rectangle (Process), Diamond (Decision).
2. Flow direction – Keep it top to bottom or left to right.
3. Connect with arrows – Show the process flow clearly.
4. One start and end – Avoid multiple starts or exits.
5. Keep it simple – Use brief and clear labels.
6. Use connectors – For complex or multi-page charts.

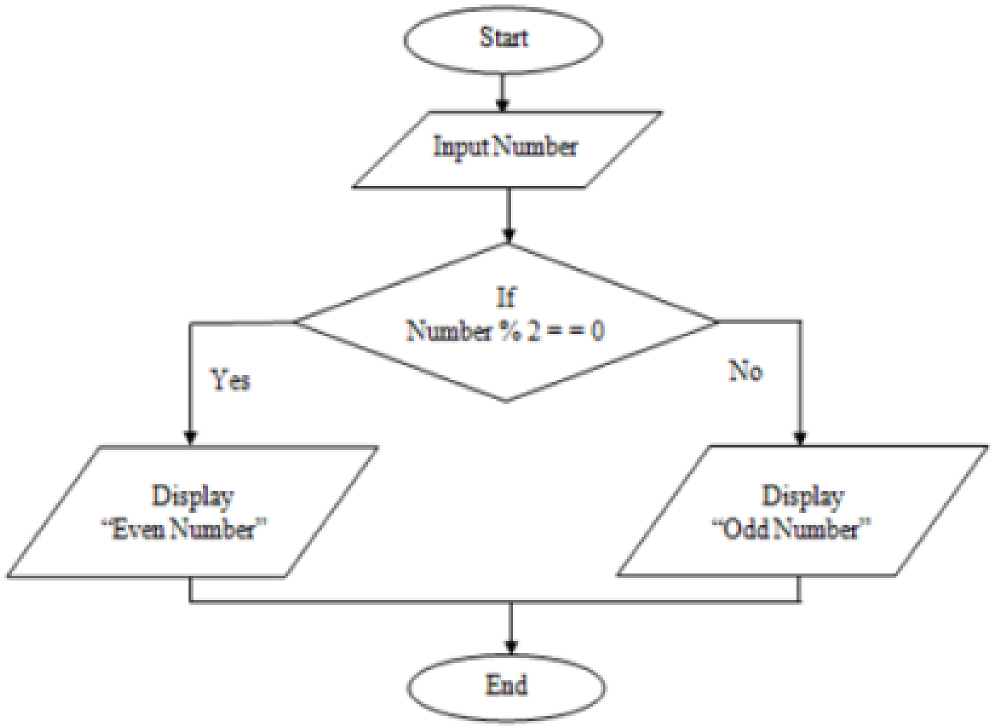
➢ Advantages of Flowcharts:

1. Easy to understand – Visual format simplifies complex processes.
2. Better communication – Clarifies tasks for teams.
3. Problem identification – Helps spot errors or inefficiencies.
4. Good for documentation – Useful for process records.
5. Supports training – Aids in teaching new users.

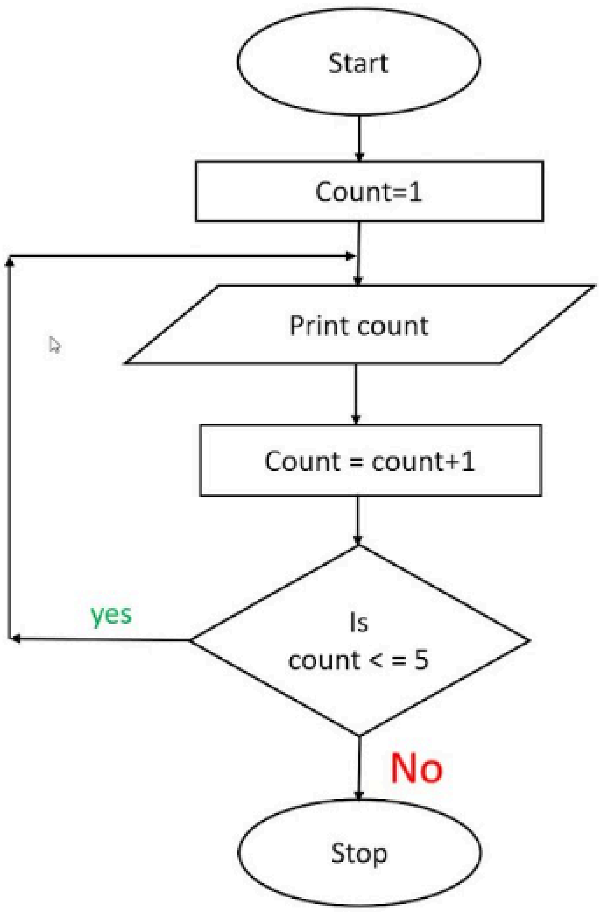
1. Draw a Flowchart to find sum of two numbers.



2. Check if a Number is Even or Odd



3. Print Numbers from 1 to 5

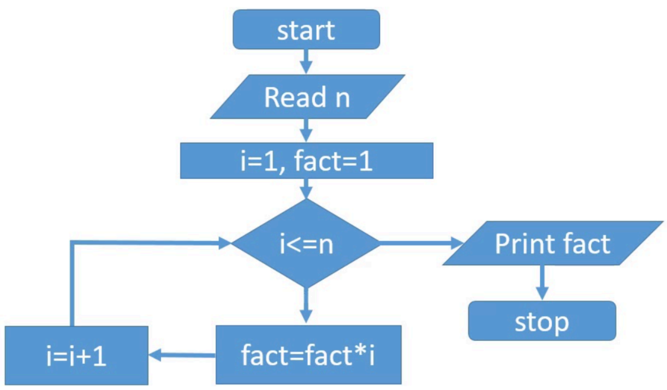


➢ Difference between Algorithm and Flowchart:

	Algorithm	**Flowchart**
Type	Written steps	Diagram / Picture
Form	Text	Visual (shapes and arrows)
Understand by	Reading	Seeing
Symbols	No symbols, only sentences	Yes, uses oval, rectangle, diamond
Example Use	Explaining logic step by step	Showing flow of process visually
Look like	Like instructions	Like a map of steps

Examples:-

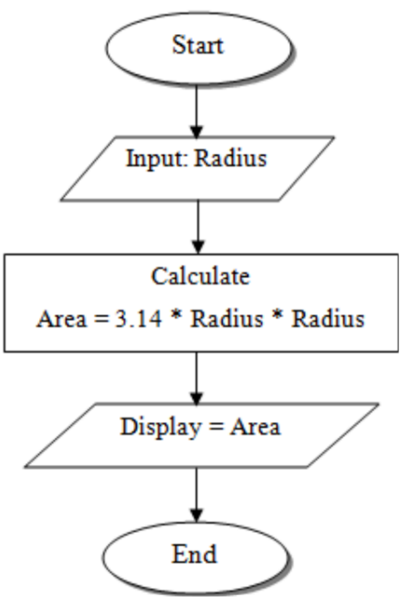
FLOWCHART FOR FACTORIAL OF A NUMBER



Algorithm: Calculate Factorial of N

Step1. Start
Step2. Read the input number N.
Step3. Initialize variables: `fact = 1`, `i = 1`.
Step4. While `i <= N`:
  a. Calculate `fact = fact * i`.
  b. Increment `i` by 1 (`i = i + 1`).
Step5. Print the value of `fact`.
Step6. Stop.

Find Area of Circle:-



Algorithm to Find Area Of Circle:-

Step1] Start
Step2] Input Radius
Step3] Calculate Area = PI * Radius * Radius
Step4] Display Area
Step5] End

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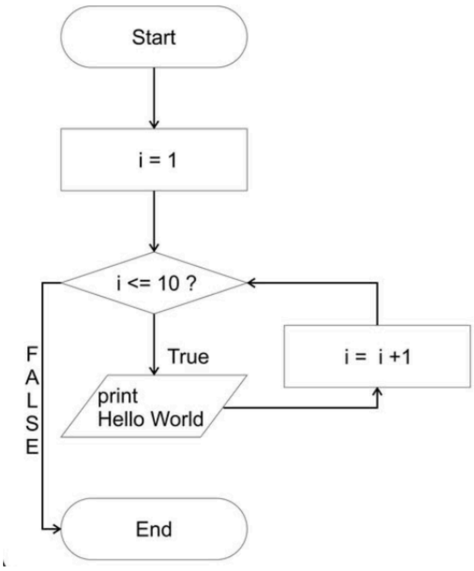
Print “Hello World” 10 times:-



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Step1:-Start

Step2:-Set a counter variable i to 1

Step3:-Repeat while i ≤ 10:

Print "Hello World"
Increment i by 1

step4:-End
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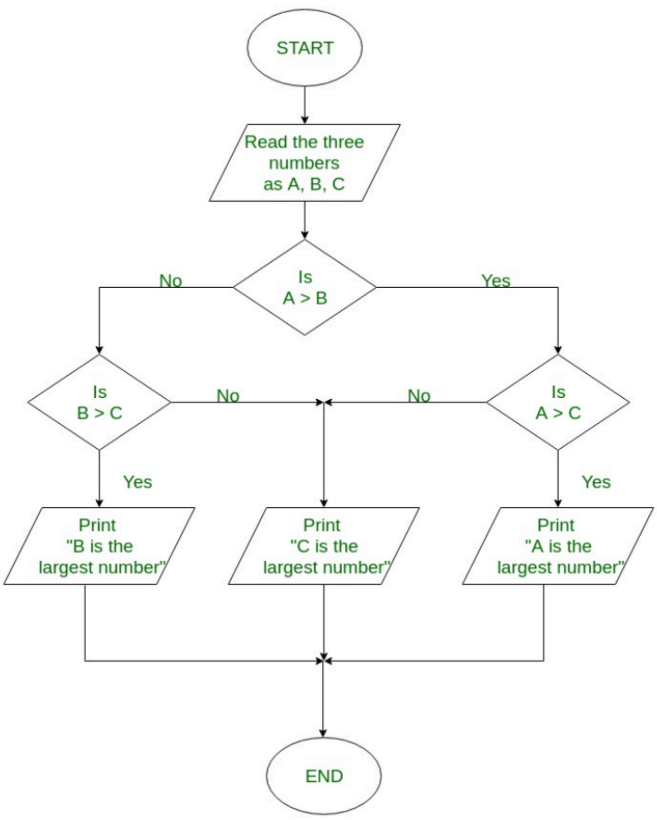
Given three numbers A, B, and C of which aim is to get the largest among these three numbers.



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Step1. Start

Step2. Read the three numbers to be compared, as A, B and C

Step3. Check if A is greater than B.

3.1 If true, then check if A is greater than C

If true, print 'A' as the greatest number
If false, print 'C' as the greatest number

3.2 If false, then check if B is greater than C

If true, print 'B' as the greatest number
If false, print 'C' as the greatest number

Step4. End
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